

is, the tighter our bounds on uniform separation rates will be. For fixed sample sizes, this condition of being balanced is always satisfied. For increasing sample sizes, the condition requires that both sample sizes increase at the same rate. In particular, under this condition, it is not possible to fix one sample size and let the other tend to infinity.

In general, we write $C_i(p_1, \dots, p_\ell)$ to express the dependence of a positive constant C_i on some parameters $p_1, \dots, p_\ell$.

We assume that we have d characteristic kernels $(x, y) \mapsto K_i(x - y)$ on $\mathbb{R} \times \mathbb{R}$ for some functions $K_i: \mathbb{R} \rightarrow \mathbb{R}$ lying in $L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ and satisfying $\int_{\mathbb{R}} K_i(u) du = 1$ for $i = 1, \dots, d$. Then, for some bandwidth $\lambda = (\lambda_1, \dots, \lambda_d) \in (0, \infty)^d$, the function¹

$$k_\lambda(x, y) := \prod_{i=1}^d \frac{1}{\lambda_i} K_i\left(\frac{x_i - y_i}{\lambda_i}\right)$$

is a characteristic kernel on $\mathbb{R}^d \times \mathbb{R}^d$ satisfying²

$$\int_{\mathbb{R}^d} k_\lambda(x, y) dx = 1 \quad \text{and} \quad \int_{\mathbb{R}^d} k_\lambda(x, y)^2 dx = \frac{\kappa_2(d)}{\lambda_1 \cdots \lambda_d} \quad (8)$$

for the constant $\kappa_2(d)$ defined later in Equation (21). Using $K_i(u) = \frac{1}{\sqrt{\pi}} \exp(-u^2)$ for $u \in \mathbb{R}$ and $i = 1, \dots, d$, for example, yields the Gaussian kernel k_λ . Using $K_i(u) = \frac{1}{2} \exp(-|u|)$ for $u \in \mathbb{R}$ and $i = 1, \dots, d$ yields the Laplace kernel k_λ . For notation purposes, given $\lambda = (\lambda_1, \dots, \lambda_d) \in (0, \infty)^d$, we also write

$$\varphi_\lambda(u) := \prod_{i=1}^d \frac{1}{\lambda_i} K_i\left(\frac{u_i}{\lambda_i}\right) \quad (9)$$

for $u \in \mathbb{R}^d$, so that $k_\lambda(x, y) = \varphi_\lambda(x - y)$ for all $x, y \in \mathbb{R}^d$. In this paper, we investigate the choice of kernel bandwidth for MMD tests. While our theoretical results on test power only hold for translation-invariant kernels on $\mathbb{R}^d \times \mathbb{R}^d$ (as introduced above), we stress that our single and aggregated MMD tests are well-defined and have well-calibrated non-asymptotic levels (Propositions 1 and 8) on any domain and for any positive definite characteristic kernel.

For clarity, we denote $\text{MMD}(p, q; \mathcal{H}_{k_\lambda})$, $\widehat{\text{MMD}}_{\mathbf{a}}^2(\mathbb{X}_m, \mathbb{Y}_n; \mathcal{H}_{k_\lambda})$, $\widehat{\text{MMD}}_{\mathbf{b}}^2(\mathbb{X}_n, \mathbb{Y}_n; \mathcal{H}_{k_\lambda})$ and h_{k_λ} (all defined in Section 2) simply by $\text{MMD}_\lambda(p, q)$, $\widehat{\text{MMD}}_{\lambda, \mathbf{a}}^2(\mathbb{X}_m, \mathbb{Y}_n)$, $\widehat{\text{MMD}}_{\lambda, \mathbf{b}}^2(\mathbb{X}_n, \mathbb{Y}_n)$ and h_λ , respectively.

When $m \neq n$, we let $\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n)$ denote the estimator $\widehat{\text{MMD}}_{\lambda, \mathbf{a}}^2(\mathbb{X}_m, \mathbb{Y}_n)$. When $m = n$, we let $\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_n, \mathbb{Y}_n)$ denote either $\widehat{\text{MMD}}_{\lambda, \mathbf{a}}^2(\mathbb{X}_n, \mathbb{Y}_n)$ or $\widehat{\text{MMD}}_{\lambda, \mathbf{b}}^2(\mathbb{X}_n, \mathbb{Y}_n)$. This means that, when $m = n$, all our results hold for both estimators.

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1. Multiplying the kernel k_λ by a positive constant C_λ does not affect the outputs of the single and aggregated tests as it simply scales both the test statistic and the quantile. The requirements that $\int_{\mathbb{R}} K_i(u) du = 1$ for $i = 1, \dots, d$ and the scaling term $(\lambda_1 \cdots \lambda_d)^{-1}$ in the definition of the kernel k_λ are not required for our theoretical results to hold. We introduce those simply for ease of notation in our statements and proofs.
 2. Detailed calculations are presented at the beginning of Appendix E.

3.2 Non-asymptotic single MMD test with a fixed bandwidth

Here, we consider the bandwidth $\lambda \in (0, \infty)^d$ to be fixed *a priori*. The null and alternative hypotheses for the two-sample problem are $\mathcal{H}_0: p = q$ against $\mathcal{H}_a: p \neq q$, or equivalently $\mathcal{H}_0: \text{MMD}_\lambda^2(p, q) = 0$ against $\mathcal{H}_a: \text{MMD}_\lambda^2(p, q) > 0$, provided that the kernels $K_1, \dots, K_d$ are characteristic. Using the samples $\mathbb{X}_m = (X_i)_{1 \leq i \leq m}$ and $\mathbb{Y}_n = (Y_j)_{1 \leq j \leq n}$, we calculate the test statistic $\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n)$. Since we want our test to be valid in the non-asymptotic framework, we cannot rely on the asymptotic distribution of $\widehat{\text{MMD}}_\lambda^2$ under the null hypothesis to compute the required threshold which guarantees the desired level $\alpha \in (0, 1)$. Instead, we use a Monte Carlo approximation to estimate the conditional $(1-\alpha)$ -quantile of the permutation-based and wild bootstrap procedures given the samples $\mathbb{X}_m$ and $\mathbb{Y}_n$ under the null hypothesis. For the estimator $\widehat{\text{MMD}}_{\lambda, a}^2(\mathbb{X}_m, \mathbb{Y}_n)$ defined in Equation (3) we use permutations, while for the estimator $\widehat{\text{MMD}}_{\lambda, b}^2(\mathbb{X}_n, \mathbb{Y}_n)$ defined in Equation (6) we use a wild bootstrap.

In Appendix B, we provide some more in-depth discussion about the relation between those two procedures. In particular, for the estimate $\widehat{\text{MMD}}_{\lambda, b}^2(\mathbb{X}_n, \mathbb{Y}_n)$, we show in Proposition 11 that using a wild bootstrap corresponds exactly to using permutations which either fix or swap X_i and Y_i for $i = 1, \dots, n$.

3.2.1 PERMUTATION APPROACH

In this case, we consider the MMD estimator defined in Equation (3) which can be written as

$$\widehat{\text{MMD}}_{\lambda, a}^2(\mathbb{X}_m, \mathbb{Y}_n) = \frac{1}{m(m-1)n(n-1)} \sum_{1 \leq i \neq i' \leq m} \sum_{1 \leq j \neq j' \leq n} h_\lambda(U_i, U_{i'}, U_{m+j}, U_{m+j'})$$

where $U_i := X_i$ and $U_{m+j} := Y_j$ for $i = 1, \dots, m$ and $j = 1, \dots, n$. Given a permutation function $\sigma: \{1, \dots, m+n\} \rightarrow \{1, \dots, m+n\}$, we can compute the MMD estimator on the permuted samples $\mathbb{X}_m^\sigma := (U_{\sigma(i)})_{1 \leq i \leq m}$ and $\mathbb{Y}_n^\sigma := (U_{\sigma(m+j)})_{1 \leq j \leq n}$ to get

$$\begin{aligned} \widehat{M}_\lambda^\sigma &:= \widehat{\text{MMD}}_{\lambda, a}^2(\mathbb{X}_m^\sigma, \mathbb{Y}_n^\sigma) \\ &= \frac{1}{m(m-1)n(n-1)} \sum_{1 \leq i \neq i' \leq m} \sum_{1 \leq j \neq j' \leq n} h_\lambda(U_{\sigma(i)}, U_{\sigma(i')}, U_{\sigma(m+j)}, U_{\sigma(m+j')}) \\ &= \frac{1}{m(m-1)} \sum_{1 \leq i \neq i' \leq m} k_\lambda(U_{\sigma(i)}, U_{\sigma(i')}) + \frac{1}{n(n-1)} \sum_{1 \leq j \neq j' \leq n} k_\lambda(U_{\sigma(m+j)}, U_{\sigma(m+j')}) \\ &\quad - \frac{2}{mn} \sum_{i=1}^m \sum_{j=1}^n k_\lambda(U_{\sigma(i)}, U_{\sigma(m+j)}). \end{aligned} \tag{10}$$

In order to estimate, with a Monte Carlo approximation, the conditional quantile of $\widehat{M}_\lambda^\sigma$ given $\mathbb{X}_m$ and $\mathbb{Y}_n$, we uniformly sample B i.i.d. permutations $\sigma^{(1)}, \dots, \sigma^{(B)}$. We denote their probability mass function by r , so that $\sigma^{(b)} \sim r$ for $b = 1, \dots, B$. We introduce the notation $\mathbb{Z}_B := (\sigma^{(b)})_{1 \leq b \leq B}$ and also simply write $\widehat{M}_\lambda^b := \widehat{M}_\lambda^{\sigma^{(b)}}$ for $b = 1, \dots, B$. We can then use the values $(\widehat{M}_\lambda^b)_{1 \leq b \leq B}$ to estimate the conditional quantile as explained in Section 3.2.3.

3.2.2 WILD BOOTSTRAP APPROACH

In this case, we assume that $m = n$ and we work with the MMD estimator $\widehat{\text{MMD}}_{\lambda, \mathbf{b}}^2$ defined in Equation (6). Recall that for this, we must assume an ordering of our samples which gives rise to a pairing (X_i, Y_i) for $i = 1, \dots, n$. Simply using permutations as presented in Section 3.2.1 would break this pairing and our estimators would consist of a signed sum of different terms because

$$\{k_\lambda(U_{\sigma(i)}, U_{\sigma(n+i)}) : i = 1, \dots, n\} \neq \{k_\lambda(U_i, U_{n+i}) : i = 1, \dots, n\}$$

for most permutations $\sigma : \{1, \dots, 2n\} \rightarrow \{1, \dots, 2n\}$. The idea is then to restrict ourselves to the permutations σ which, for $i = 1, \dots, n$, either fix or swap X_i and Y_i , in the sense that $\{U_{\sigma(i)}, U_{\sigma(n+i)}\} = \{U_i, U_{n+i}\}$, so that $k_\lambda(U_{\sigma(i)}, U_{\sigma(n+i)}) = k_\lambda(U_i, U_{n+i})$. We show in Proposition 11 in Appendix B that this corresponds exactly to using a wild bootstrap, which we now define.

Given n i.i.d. Rademacher random variables $\epsilon := (\epsilon_1, \dots, \epsilon_n)$ with values in $\{-1, 1\}^n$, we let

$$\widehat{M}_\lambda^\epsilon := \frac{1}{n(n-1)} \sum_{1 \leq i \neq j \leq n} \epsilon_i \epsilon_j h_\lambda(X_i, X_j, Y_i, Y_j). \quad (11)$$

As for the permutation approach, in order to obtain a Monte Carlo estimate of the conditional quantile of $\widehat{M}_\lambda^\epsilon$ given $\mathbb{X}_n$ and $\mathbb{Y}_n$, for $b = 1, \dots, B$, we generate n i.i.d. Rademacher random variables $\epsilon^{(b)} := (\epsilon_1^{(b)}, \dots, \epsilon_n^{(b)})$ with values in $\{-1, 1\}^n$ and compute $\widehat{M}_\lambda^b := \widehat{M}_\lambda^{\epsilon^{(b)}}$. We write $\mathbb{Z}_B := (\epsilon^{(b)})_{1 \leq b \leq B}$ and denote their probability mass function as r to be consistent with the notation introduced in Section 3.2.1, so that $\epsilon^{(b)} \sim r$ for $b = 1, \dots, B$. We next show in Section 3.2.3 how to estimate the conditional quantile using $(\widehat{M}_\lambda^b)_{1 \leq b \leq B}$.

3.2.3 SINGLE MMD TEST: DEFINITION AND LEVEL

Depending on which MMD estimator we use, either $\widehat{\text{MMD}}_{\lambda, \mathbf{a}}^2$ from Equation (3) or $\widehat{\text{MMD}}_{\lambda, \mathbf{b}}^2$ from Equation (6), we obtain $(\widehat{M}_\lambda^b)_{1 \leq b \leq B}$ either as in Section 3.2.1 or as in Section 3.2.2, respectively. Inspired by the work of Romano and Wolf (2005a, Lemma 1) and Albert et al. (2022), in order to obtain the prescribed non-asymptotic test level, we also add the original MMD statistic

$$\widehat{M}_\lambda^{B+1} := \widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n)$$

which corresponds either to the case where the permutation is the identity or where the n Rademacher random variables are equal to 1. We can then estimate the conditional quantile of the distribution of either $\widehat{M}_\lambda^\sigma$ or $\widehat{M}_\lambda^\epsilon$ given $\mathbb{X}_m$ and $\mathbb{Y}_n$ under the null hypothesis $\mathcal{H}_0 : p = q$ by using a Monte Carlo approximation. In particular, our estimator of the conditional $(1 - \alpha)$ -quantile is given by

$$\widehat{q}_{1-\alpha}^{\lambda, B}(\mathbb{Z}_B | \mathbb{X}_m, \mathbb{Y}_n) := \inf \left\{ u \in \mathbb{R} : 1 - \alpha \leq \frac{1}{B+1} \sum_{b=1}^{B+1} \mathbb{1}(\widehat{M}_\lambda^b \leq u) \right\} = \widehat{M}_\lambda^{\bullet[(B+1)(1-\alpha)]} \quad (12)$$

where $\widehat{M}_\lambda^{\bullet 1} \leq \dots \leq \widehat{M}_\lambda^{\bullet B+1}$ denote the ordered simulated test statistics $(\widehat{M}_\lambda^b)_{1 \leq b \leq B+1}$. We then define the single MMD test $\Delta_\alpha^{\lambda, B}$ for some given bandwidth $\lambda \in (0, \infty)^d$ as

$$\Delta_\alpha^{\lambda, B}(\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_B) := \mathbb{1} \left(\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) > \widehat{q}_{1-\alpha}^{\lambda, B}(\mathbb{Z}_B | \mathbb{X}_m, \mathbb{Y}_n) \right).$$